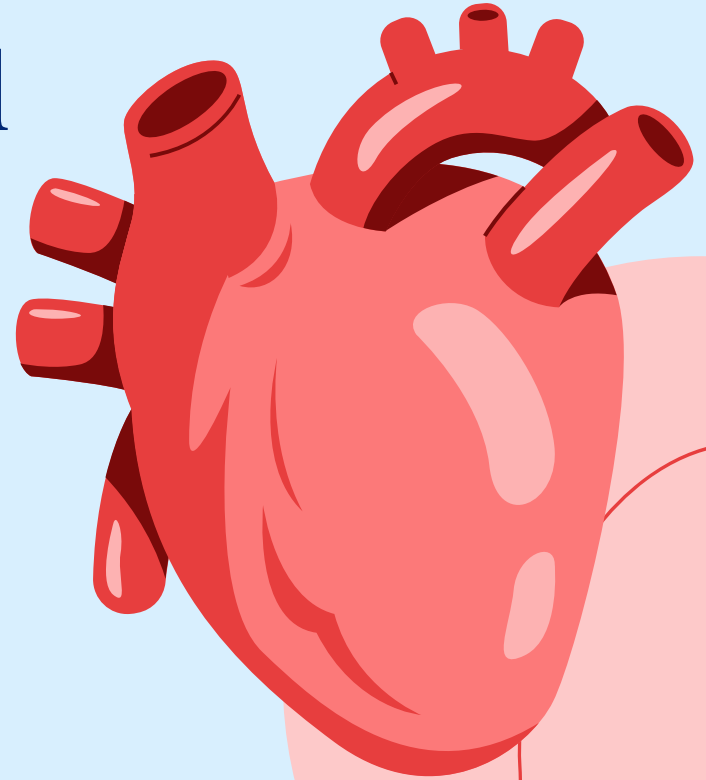


Enhancing Artificial Heart Valve Testing & Efficacy

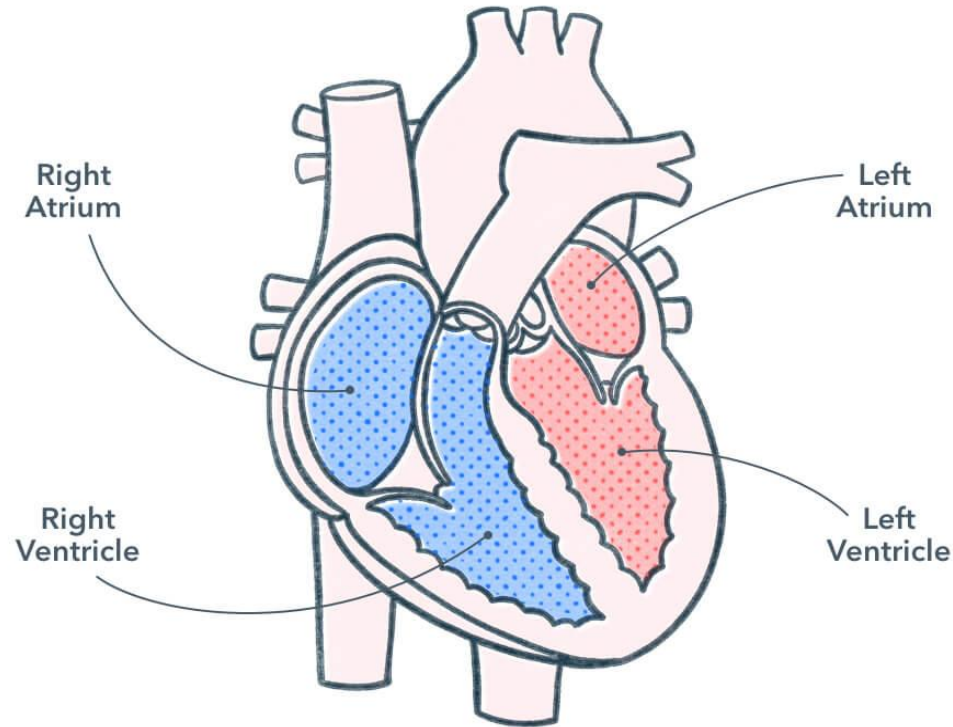
By: Arian Salari

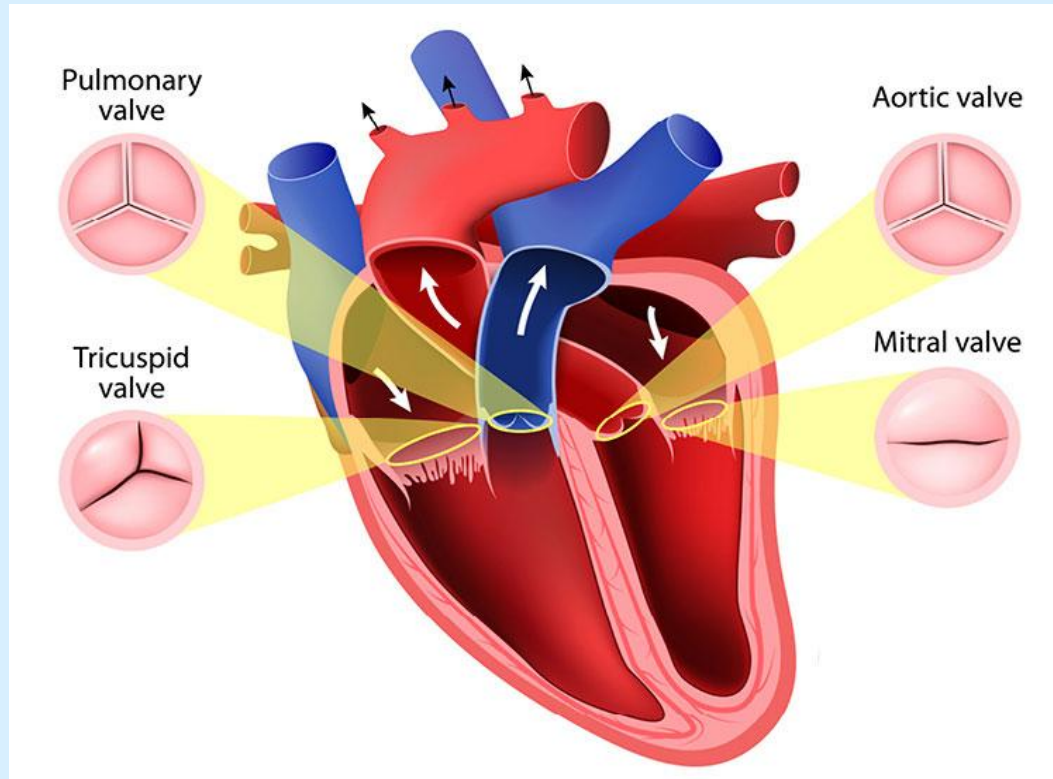
Irvine Valley College





Basic Anatomical Overview of the Heart





Heart valves are supposed to ensure proper blood flow within the heart.

What Are Artificial Heart Valves?



Mechanical Valve



Biological Valve

Medical device designed to replace damaged or dysfunctional heart valves. They help restore heart valve function in patients with defective valves which would improve their quality of life.



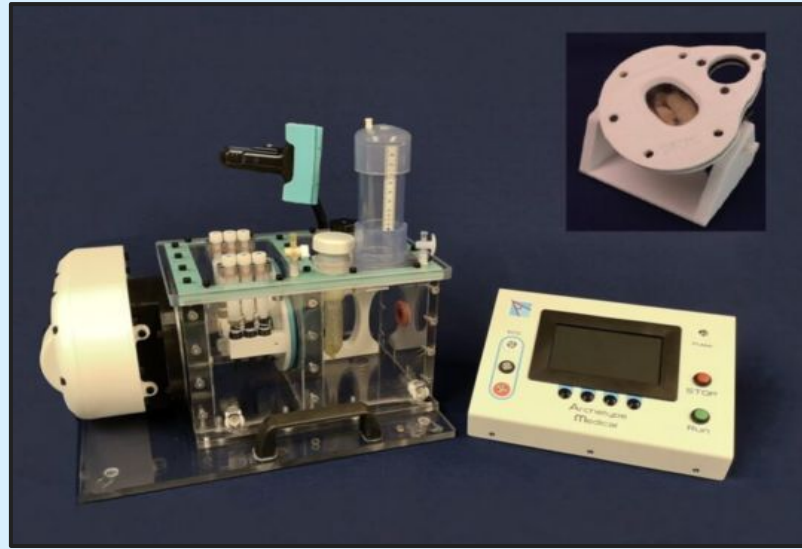
How Are Artificial Valves Tested?

Different Methods:

- **Mechanical heart simulations**: Machine replication of functioning human heart; consistent testing; controlled and repeatable conditions.
- **Animal tests**: Animals like pigs are used to observe how artificial heart valves perform in living organisms.
- **Computer simulations and models**: Enable testing without needing physical prototypes.



♥ Challenges With Mechanical Testing



Mechanical tests: require costly and specialized mechanical equipment which still don't end up fully replicating the human hearts dynamics and anatomy. This leads to additional testing rounds, increased costs, and longer development times.

Challenges With Animal Testing



Animal studies: tend to get expensive and are ethically questionable. Physiological and anatomical differences between animals and humans also mean results may not always predict human outcomes.

♥ Challenges With Computer Testing



Developing accurate **computational simulations** is resource-intensive and can't yet model all biological complexities. This leads to iterative adjustments and increased costs, further delaying testing timelines.

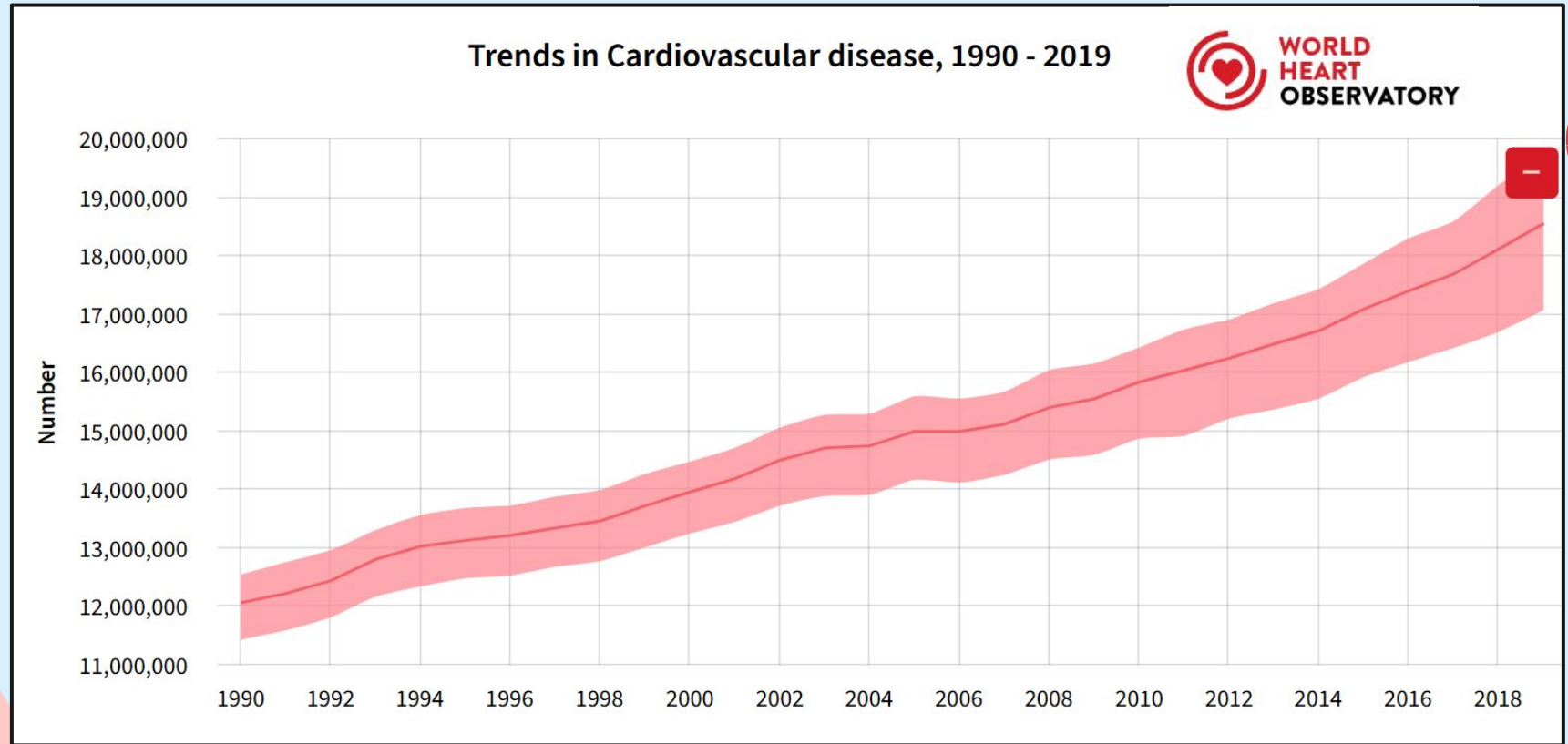


Why We Need Improved Testing

1. **Heart Valve Replication:** Captures the heart's full range of functions and structures, improving test accuracy.
2. **Reduced Device Failure:** Lowers risk of failure after approval, ensuring safer outcomes for patients.
3. **Company Benefits:** Streamlines development, saving time and reducing production costs.
4. **Patient Access:** Enables faster availability of safer, more durable heart valves at a lower cost.



Heart Disease is the number one cause of death in the US and its still on the rise so this should be prioritized.



Graph from World Heart Federation



What We Researched

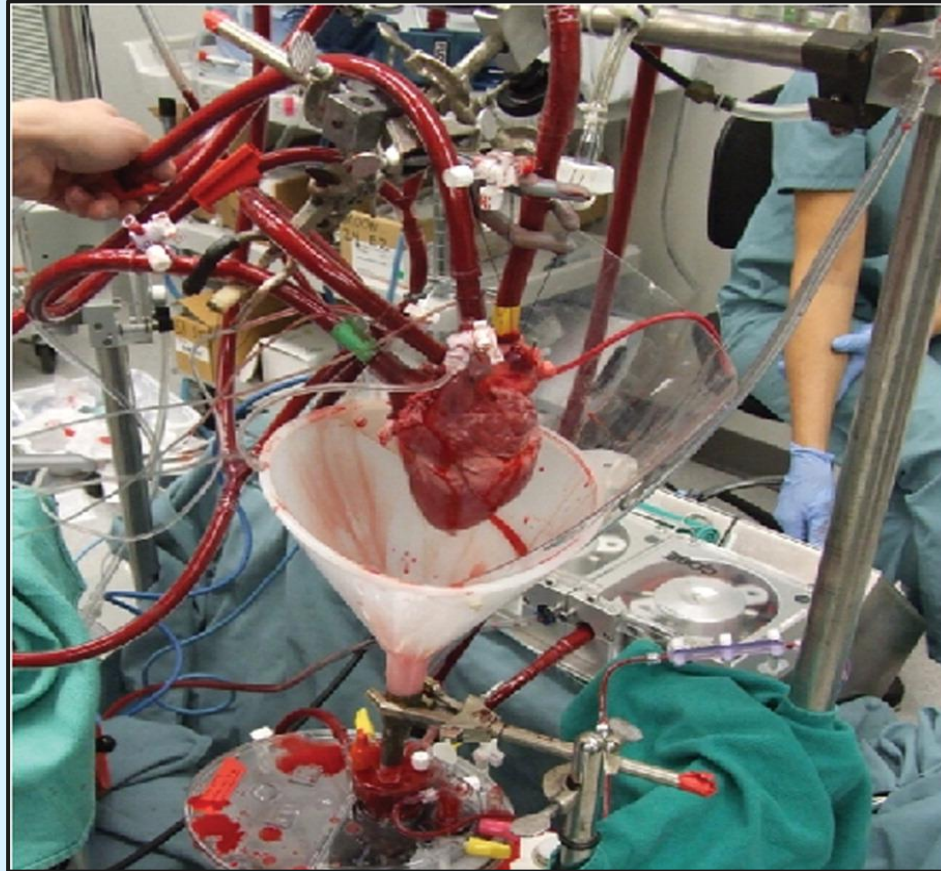
1. **Foundation in BrainEx Technology:** Builds on a Yale-led project that used technology to restore circulation and cellular functions in a deceased pig's brain.
2. **Research Adaptation:** Aims to apply this technology to slow cellular and organ deterioration in a pig's heart immediately after the final heartbeat.
3. **Direct Heart Valve Testing:** Slowing deterioration would allow for the preserved pig heart to be used directly for testing artificial heart valves, closely simulating human conditions.



What is NEHP and How Does it Work?

1. **Acronym of NEHP:** Stands for Normothermic Ex-Vivo Heart Perfusion.
2. **Process Function:** Circulates oxygenated blood and essential nutrients through the heart at body temperature.
3. **Deterioration Prevention:** Slows post-death deterioration caused by lack of blood flow, oxygen, and nutrients.
4. **Cellular Preservation:** Delays cellular breakdown, increasing the heart's viability as a live testing platform.
5. **Enhanced Testing Capability:** Enables the use of perfused hearts for more realistic and reliable testing.

Perfused Heart





Steps of the Perfusion Process

1. **Harvesting and Preparation:** Heart is carefully removed and flushed with a preservation solution to slow cell damage.
2. **Rewarming and Perfusion Start:** Heart is warmed to body temperature and attached to NEHP machine which supplies oxygenated blood through the heart.
3. **Continuous Monitoring:** Heart rate, pressure, and other vitals are monitored and adjusted to maintain optimal conditions.
4. **Extended Preservation:** NEHP preserves heart viability for several hours, enhancing transplant success rates.





Advancing Heart Preservation with NEHP

(Current Method); Cold Static Storage (CSS):

- Current gold standard for heart preservation is **6 hours**.
- Commonly used for simplicity and cost-effectiveness.

(New); Normothermic Ex Vivo Heart Perfusion (NEHP):

- Extends preservation time beyond CSS.
- **Study Results (Zhao et al., 2024):**
 - **Sample:** 27 pig hearts tested with NEHP
 - **Outcomes:** All 27 hearts survived **up to 24 hours**, showcasing improved durability and longevity of this method.

Electrically Stimulating a Perfused Heart for Simulated Physiology

1. **External stimulator** connects to the **perfused heart** to simulate a **natural heartbeat**.
2. **Electrodes** are placed on the **atria and ventricles** to mimic natural heart contractions.
3. Stimulator **controls heart rate**, keeping it within a normal range of **60-100 BPM**.
4. Allows for **safe and accurate testing** of artificial heart valves in **realistic conditions**.



Advantages of Using a NEHP Perfused Pig Heart

1. **Realistic Conditions:** Maintains body temperature and replicates human physiology.
2. **Extended Testing Duration:** The prolonged testing period allows for better evaluation of valve durability and performance.
3. **Good Functional Assessment:** Real-time monitoring of valve response to varied pressures and heart rates in an actual heart under controlled conditions.
4. **Reduced Animal Testing:** Minimizes reliance on live animals which addresses ethical concerns.





Mitigating Ethical Concerns

Minimizes Animal Use: The NEHP method provides a functional, realistic heart environment, reducing the need for live animal testing. By using NEHP, researchers can gather accurate data while adhering to ethical standards.

Fun Fact: “Over 110 million animals die in U.S. laboratories used for animal testing every year” (Diaz 1).





More Benefits

- **Extended Heart Preservation:** NEHP maintains heart viability for hours, reducing the frequency of new samples needed and lowering costs.
- **Streamlined Testing Process:** Real-time monitoring and controlled conditions allow for faster data collection and more efficient testing cycles.
- **Reduced Animal Procurement Costs:** Less reliance on animal models means fewer expenses related to animal care, handling, and ethical compliance.
- **Enhanced Accuracy for Fewer Defects:** NEHP's realistic testing environment filters out defective valves early, reducing long-term hospital recalls and associated costs.
- **Increased Patient Access:** Cost-effective development leads to more affordable valves, expanding access for patients.





Summary

Our research focuses on enhancing the accuracy of artificial heart valve testing through a heart reviving method. By connecting electrodes to a NEHP perfused heart, the heart's rhythm and rate are simulated recreating body-like conditions that can be controlled. This approach enables reliable testing ultimately improving the efficacy of artificial heart valves for patients.

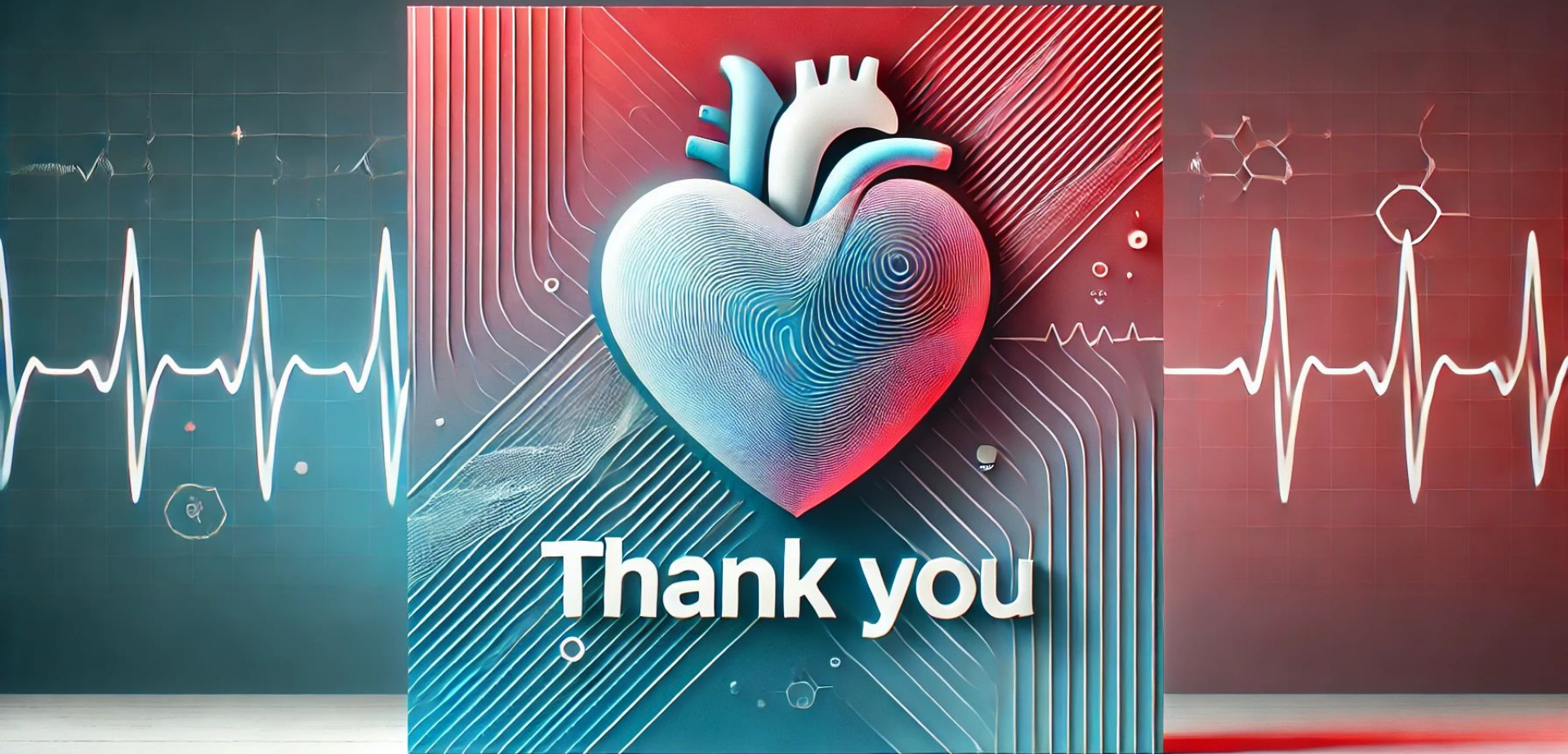




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Thank you